



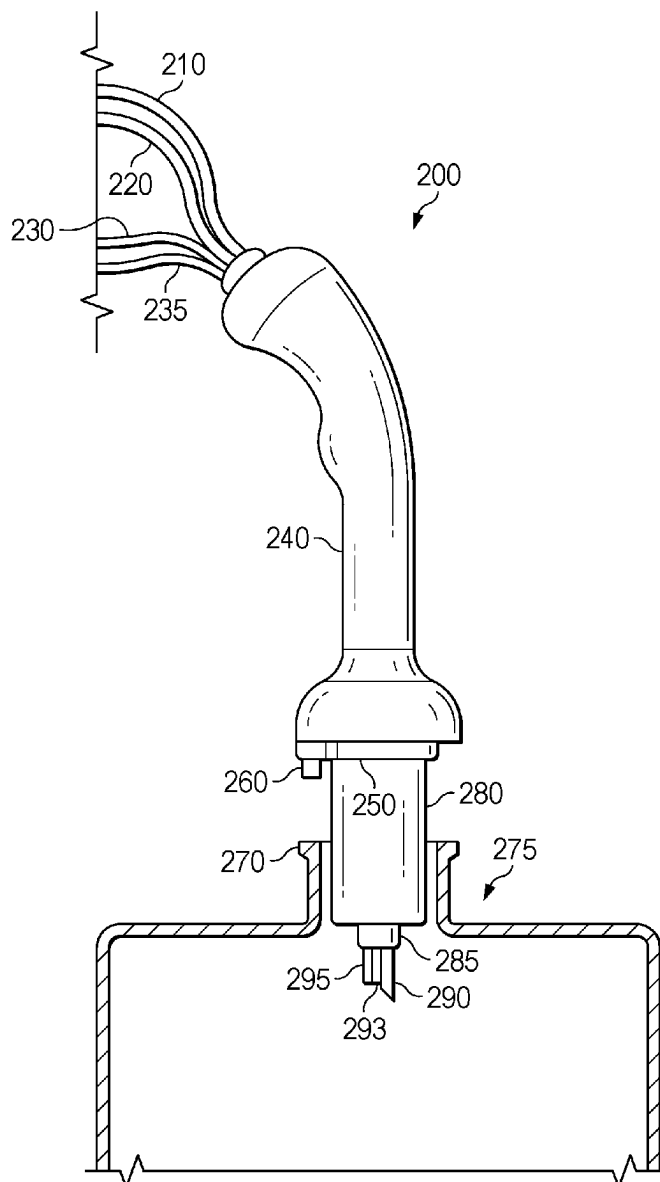
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Greene et al.(10) **Pub. No.: US 2025/0286159 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SYSTEMATIC AUTO WATERING (SAW)**(71) Applicant: **The Boeing Company**, Arlington, VA
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John M. Holder, Plano, TX (US)(21) Appl. No.: **18/601,974**(22) Filed: **Mar. 11, 2024****Publication Classification**(51) **Int. Cl.**
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(57)

ABSTRACT

A systematic automatic watering system tool is described. In one embodiment, a systematic automatic battery watering system tool includes a non-contact electrolyte level sensing probe. The probe includes an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port, an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer adapted to stop a flow of liquid into the battery cell in response to detecting a strength of a received signal reflected by a surface of an electrolyte in the battery cell exceeding a received signal strength threshold. Initially developed to water battery cells, its design versatility extends to various other applications.



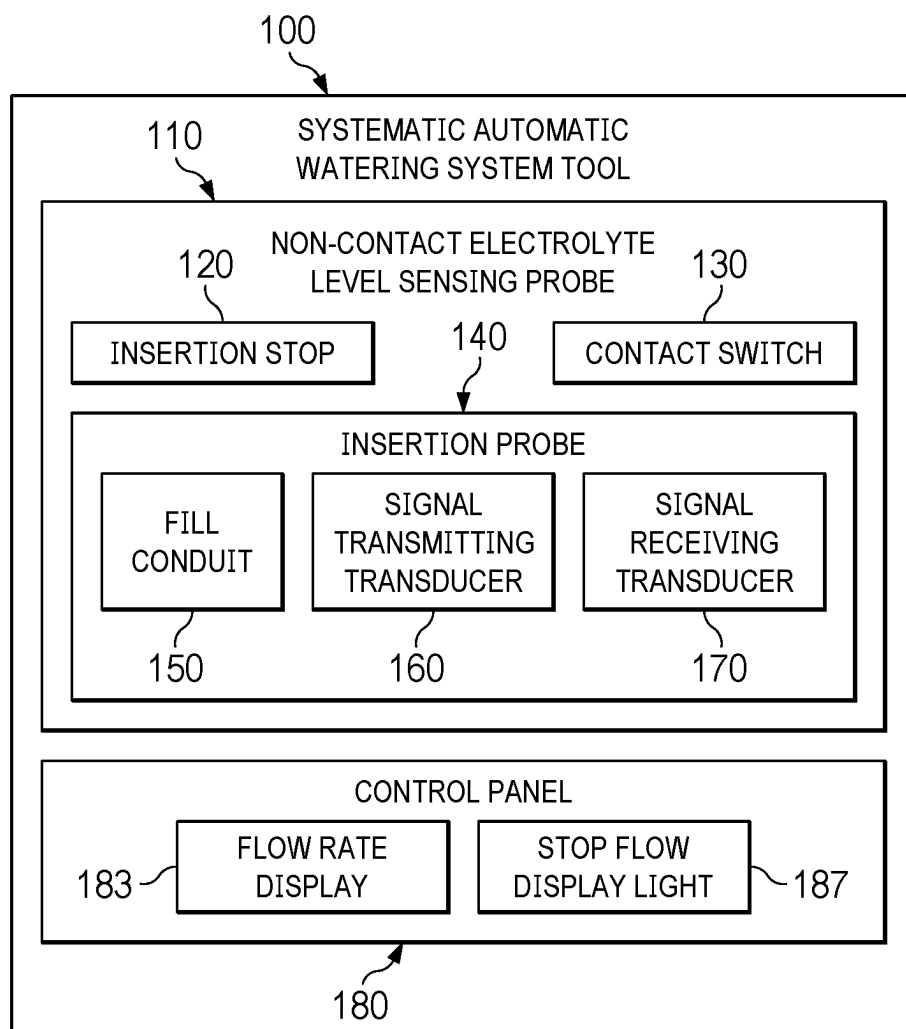


FIG. 1

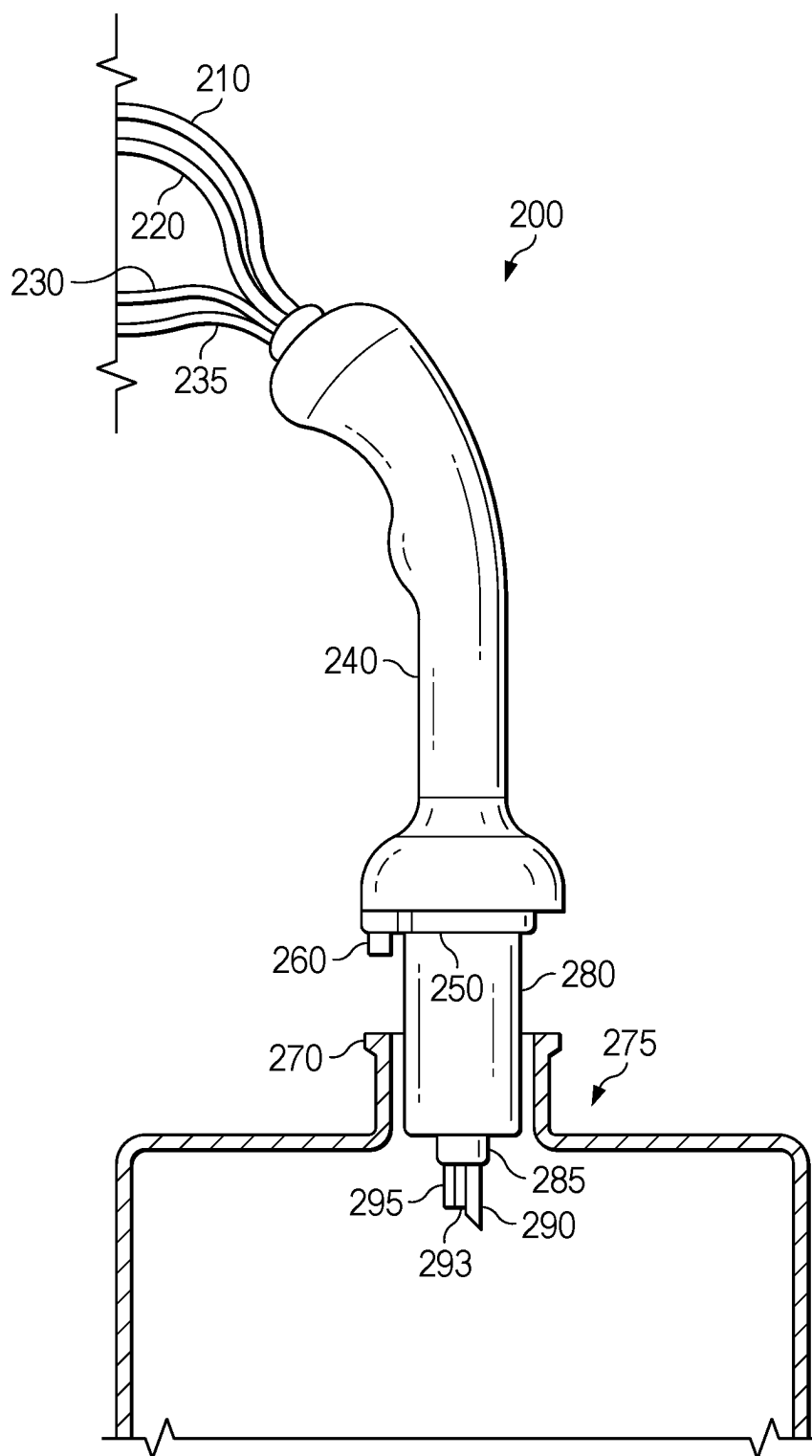


FIG. 2

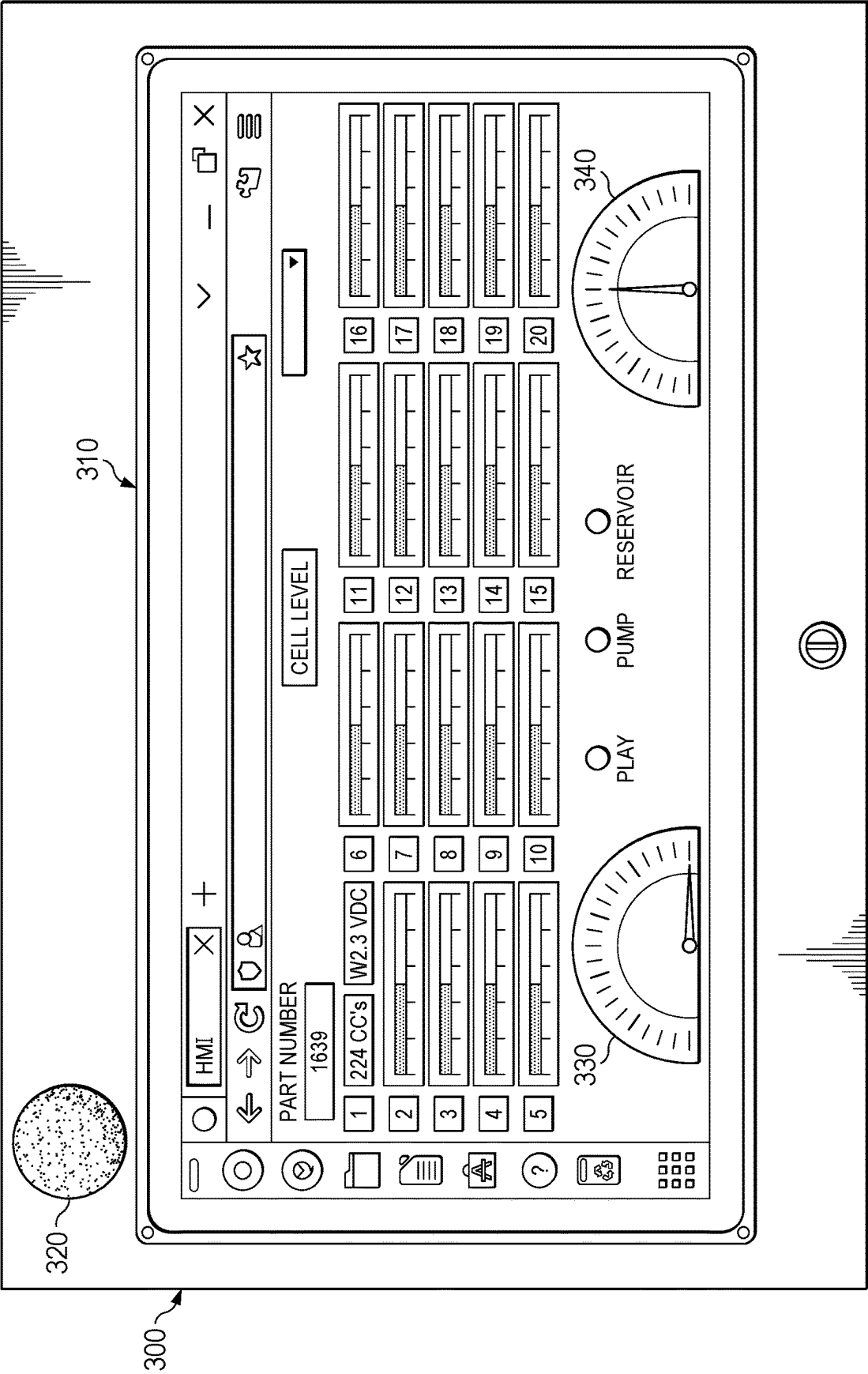


FIG. 3

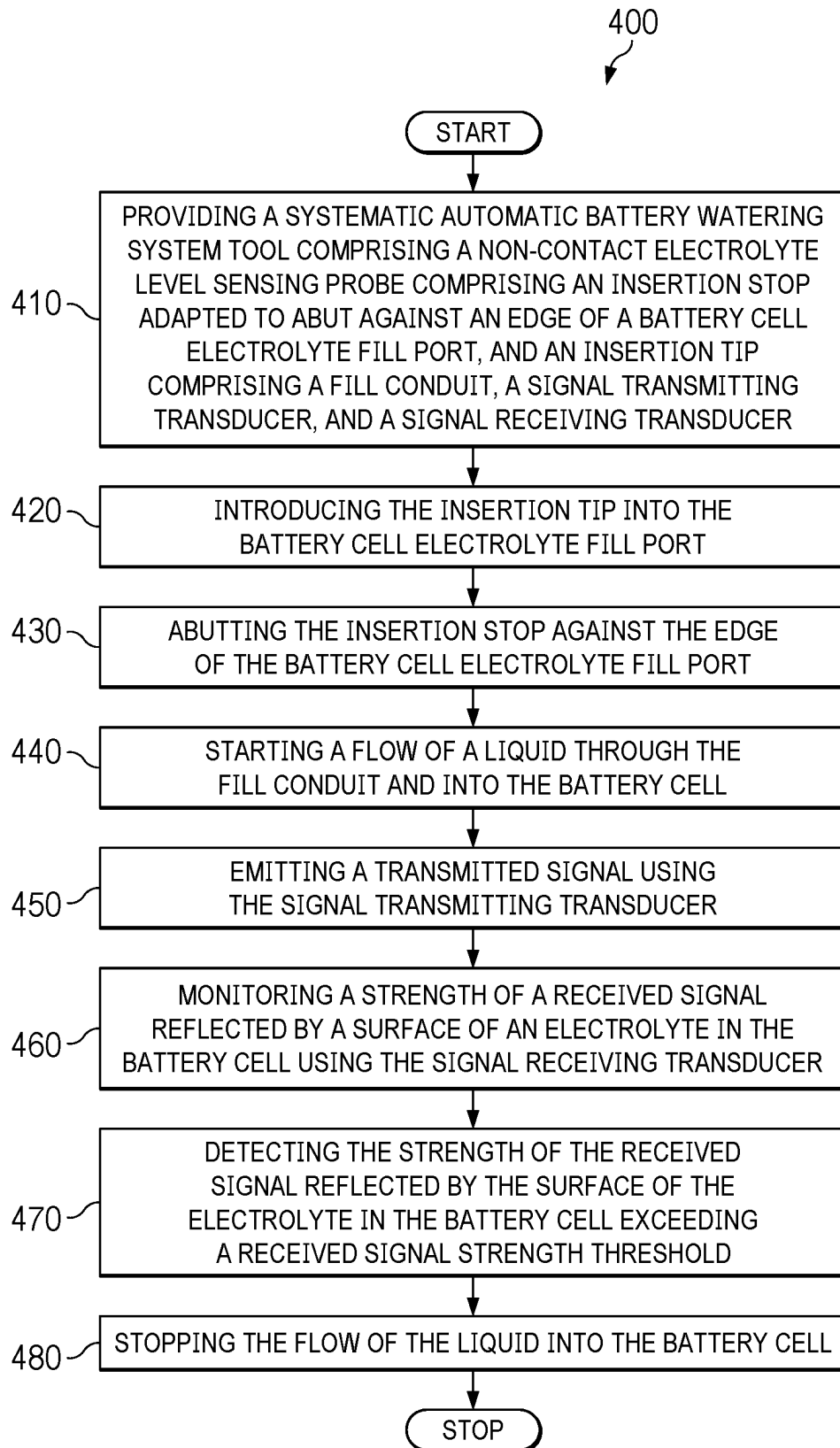


FIG. 4

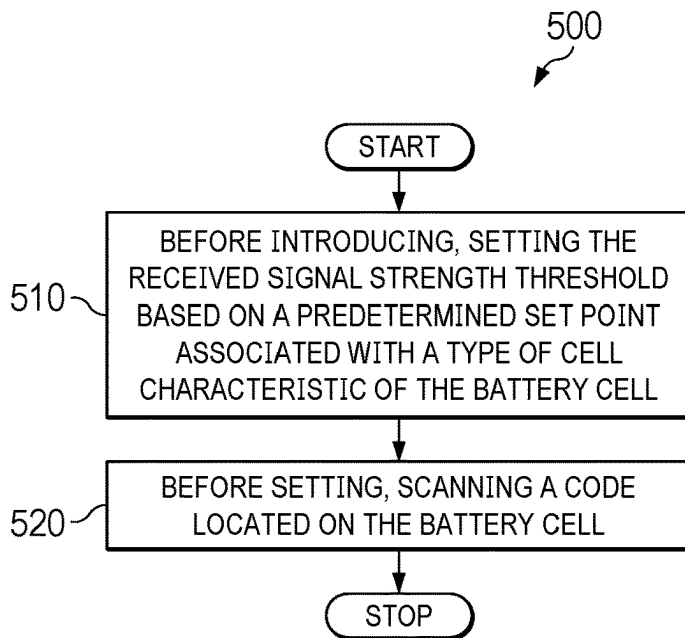


FIG. 5

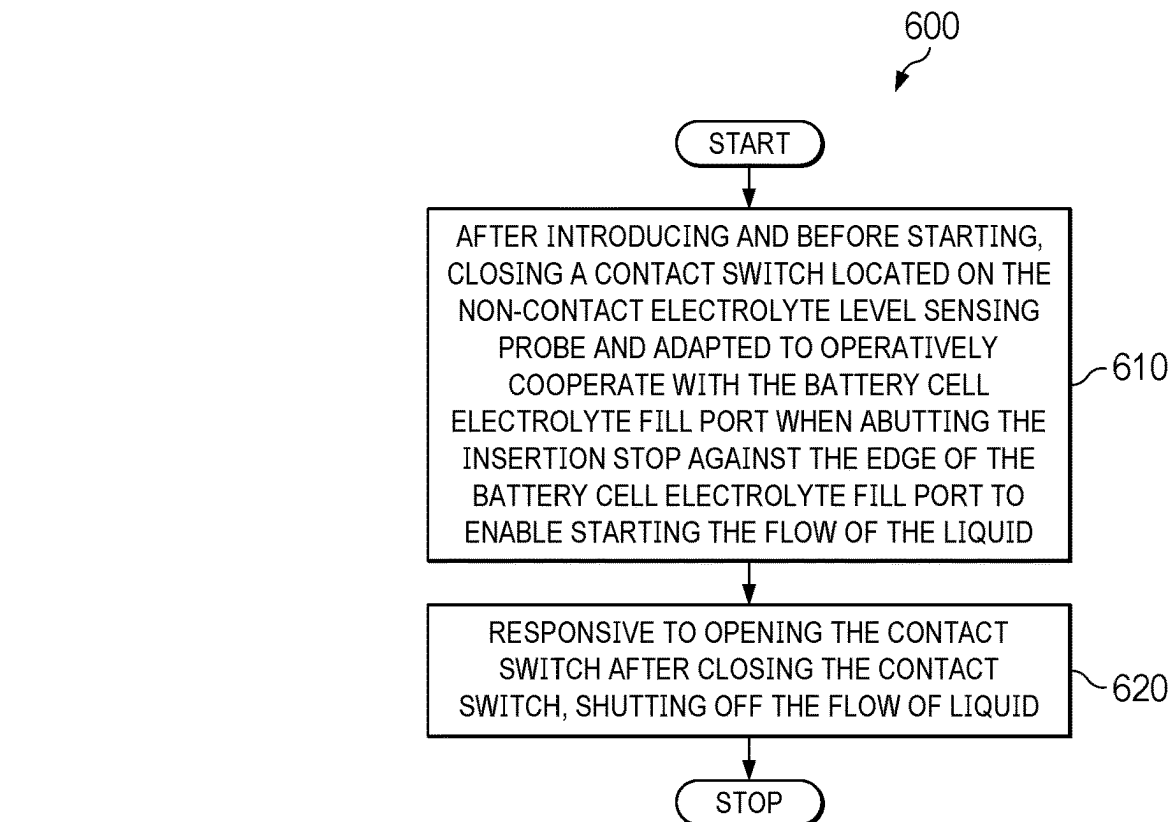


FIG. 6

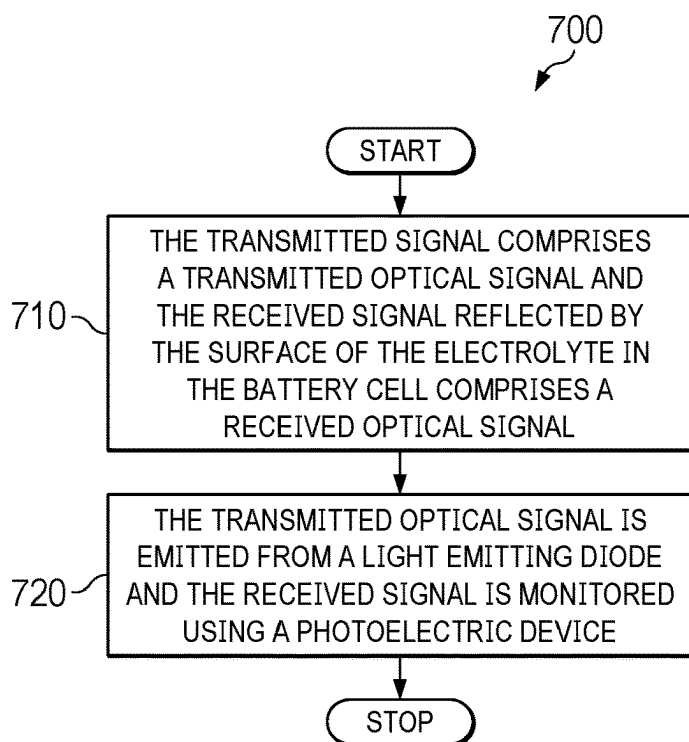


FIG. 7

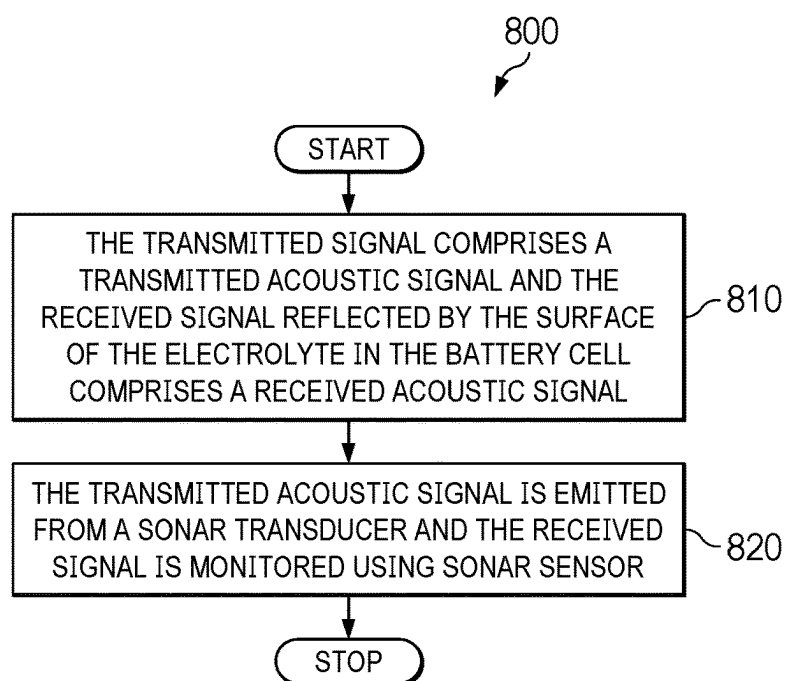


FIG. 8

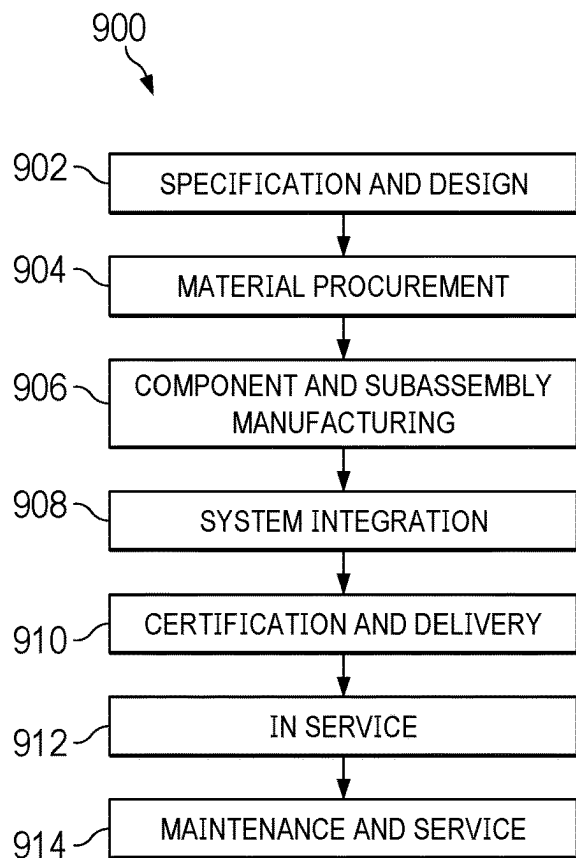


FIG. 9

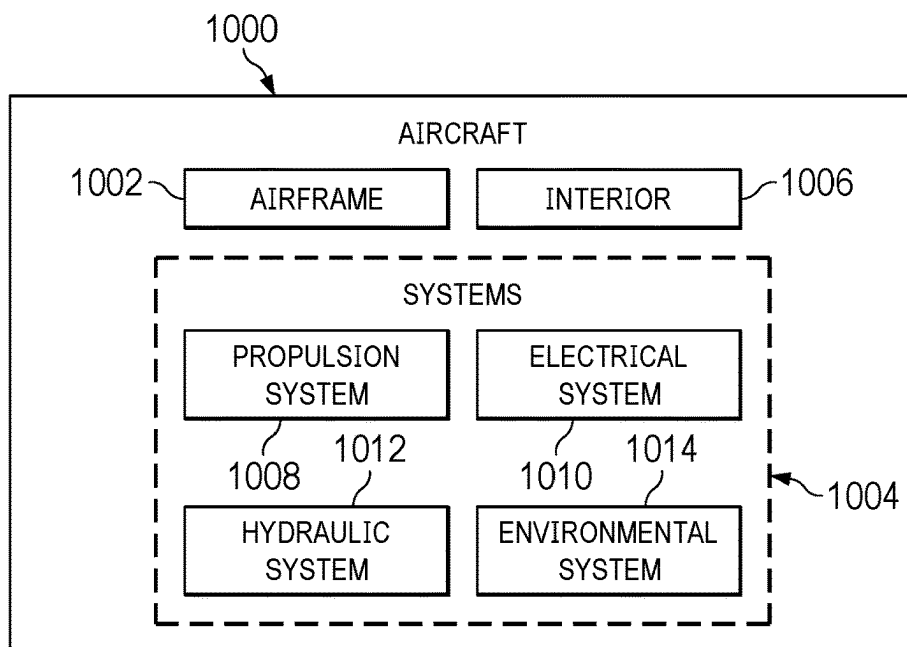


FIG. 10

SYSTEMATIC AUTO WATERING (SAW)

BACKGROUND INFORMATION

1. Field

[0001] The present disclosure relates generally to a systematic automatic watering system and systematic automatic watering system methodologies, and more specifically to a systematic automatic watering tool for battery cell electrolyte maintenance and associated methodologies.

2. Background

[0002] Vehicles are dependent on their batteries. Batteries deteriorate over time and need to be serviced according to manufacturer procedures in order to extend their useful lifetimes. One of the aspects of battery service is maintaining electrolyte level within specifications in each of the cells of a battery.

[0003] Typically, the battery manufacturer procedure calls for a time-consuming filling process with use of a syringe to maintain electrolyte level. Drawbacks to this approach are lack of precision and unpredictable productivity. Other competitor solutions are very inaccurate and miss aspects of compliance.

[0004] One previous approach to maintaining electrolyte level within each of the cells of a battery uses a vacuum pump and a pressure sensor. During watering when liquid is added to raise the electrolyte level, the rising electrolyte eventually touches a vacuum system orifice causing the pressure sensor to detect a pressure drop. Consequently, a relay can be used to shut off the flow of water.

[0005] However, a significant drawback to this previous approach is that before every shut-off some corrosive electrolyte is drawn into the vacuum system through the orifice. The problems inherent with this previous approach are related to corrosion and inconsistency. Because a vacuum is used to make contact with rising electrolyte, corrosive potassium hydroxide electrolyte is introduced to the internals of the system. This causes false pressure reading over time leading to inconsistency. Eventually, corrosion degrades the vacuum system, especially the conduit defining the orifice, leading to absence of precision and even contamination of the electrolyte. Ultimately, systems that implement this previous approach inevitably fail before the end of their theoretical life cycle. Another significant drawback to this contact approach is cross-contamination of electrolyte between cells.

[0006] Therefore, it would be desirable to have a watering system tool, as well as methods of using that tool that take into account at least some of the issues discussed above, as well as other possible issues.

SUMMARY

[0007] There is a need for the following embodiments of the present disclosure. Of course, the present disclosure is not limited to these embodiments.

[0008] Embodiments of the present disclosure can provide a systematic non-contact automatic battery watering system tool and methodology. No corrosive electrolytes or other solutions enter the system alleviating failure problems and providing a significant commercial advantage.

[0009] An embodiment of the present disclosure provides a method of automatically watering a battery cell, compris-

ing: providing a systematic automatic battery watering system tool comprising a non-contact electrolyte level sensing probe comprising an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port, and an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer; introducing the insertion tip into the battery cell electrolyte fill port; abutting the insertion stop against the edge of the battery cell electrolyte fill port; starting a flow of a liquid through the fill conduit and into the battery cell; emitting a transmitted signal using the signal transmitting transducer; monitoring a strength of a received signal reflected by a surface of an electrolyte in the battery cell using the signal receiving transducer; detecting the strength of the received signal reflected by the surface of the electrolyte in the battery cell exceeding a received signal strength threshold; and stopping the flow of the liquid into the battery cell.

[0010] Another embodiment of the present disclosure provides an apparatus for automatically watering a battery cell, comprising: a control panel; a non-contact electrolyte level sensing probe coupled to the control panel, the non-contact electrolyte level sensing probe comprising: an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port; an insertion tip comprising a fill conduit; a signal transmitting transducer; and a signal receiving transducer adapted to stop a flow of liquid into the battery cell in response to detecting a strength of a received signal reflected by a surface of an electrolyte in the battery cell exceeding a received signal strength threshold.

[0011] Another embodiment of the present disclosure provides a method of automatically watering, comprising: providing a systematic automatic watering system tool comprising a non-contact liquid level sensing probe comprising an insertion stop adapted to abut against an edge of a cell fill port, and an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer; setting a received signal strength threshold based on a predetermined set point associated with a type of cell characteristic of a cell; introducing the insertion tip into the cell fill port of the cell; closing a contact switch located on the non-contact liquid level sensing probe and adapted to operatively cooperate with the cell fill port when abutting the insertion stop against the edge of the cell fill port to enable starting a flow of a liquid; abutting the insertion stop against the edge of the cell fill port; starting a flow of a liquid through the fill conduit and into the cell; emitting a transmitted signal using the signal transmitting transducer; monitoring a strength of a received signal reflected by a surface of a liquid in the cell using the signal receiving transducer; detecting the strength of the received signal reflected by the surface of the liquid in the cell exceeding the received signal strength threshold; and stopping the flow of the liquid into the cell.

[0012] The features and functions can be achieved independently in various embodiments of the present disclosure or may be combined in yet other embodiments in which further details can be seen with reference to the following description and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The novel features believed characteristic of the illustrative embodiments are set forth in the appended claims. The illustrative embodiments, however, as well as a preferred mode of use, further objectives and features

thereof, will best be understood by reference to the following detailed description of an illustrative embodiment of the present disclosure when read in conjunction with the accompanying drawings, wherein:

[0014] FIG. 1 is an illustration of a block diagram of an apparatus comprising a systematic automatic watering system in accordance with an illustrative embodiment;

[0015] FIG. 2 is an illustration of a non-contact electrolyte level sensing probe for a systematic automatic watering system in accordance with an illustrative embodiment;

[0016] FIG. 3 is an illustration of a control panel for a systematic automatic watering system in accordance with an illustrative embodiment;

[0017] FIG. 4 is an illustration of a flowchart of a process for systematic automatic battery watering in accordance with an illustrative embodiment;

[0018] FIG. 5 is an illustration of a flowchart of a process for scanning a code located on a battery cell and setting a received signal strength threshold based on a predetermined set point associated with a type of cell characteristic of the battery cell in accordance with an illustrative embodiment;

[0019] FIG. 6 is an illustration of a flowchart of a process for closing a contact switch, opening the contact switch, and shutting off a flow of liquid in accordance with an illustrative embodiment;

[0020] FIG. 7 is an illustration of a flowchart of a process for monitoring a received signal using a photoelectric device in accordance with an illustrative embodiment;

[0021] FIG. 8 is an illustration of a flowchart of a process for monitoring a received signal using a sonar sensor in accordance with an illustrative embodiment;

[0022] FIG. 9 is an illustration of an aircraft manufacturing and service method in a form of a block diagram in accordance with an illustrative embodiment; and

[0023] FIG. 10 is an illustration of an aircraft in a form of a block diagram in which an illustrative embodiment may be implemented.

DETAILED DESCRIPTION

[0024] Embodiments presented in the present disclosure and the various features and advantageous details thereof are explained more fully with reference to the nonlimiting embodiments that are illustrated in the accompanying drawings and detailed in the following description. Descriptions of well-known materials, techniques, components and equipment are omitted so as not to unnecessarily obscure the embodiments of the present disclosure in detail. It should be understood, however, that the detailed description and the specific examples are given by way of illustration only and not by way of limitation. Various substitutions, modifications, additions and/or rearrangements within the scope of the underlying inventive concept will become apparent to those skilled in the art from this disclosure.

[0025] An embodiment of this disclosure includes a systematic automatic watering system tool and methodology. In this embodiment, a readily commercially available industrial PLC (programmable logic controller) can be utilized for the logic together with a readily commercially available lab accurate flow sensor, and combined with multiple level sensing guns (level sensing probes) corresponding to multiple battery cell form factors. The PLC programing can adapt to differences in batteries and automatically store cell watering characterization data for compliance. The indi-

vidual cells if provided with bar codes can be scanned by the program ensuring no user error when setting up the system for use.

[0026] Commercially available PLCs are updatable in the future to allow continued use in watering batteries, no matter the changes made to the cells of the batteries by manufacturers. No corrosive solutions enter the system alleviating failure problems and providing a significant commercial advantage.

[0027] In a first example, an embodiment of the systematic automatic watering system uses a non-contact laser to reflect off the top of the water in an electrolyte. The analog output is based off the intensity of the diffused light. The use of a photodetector in combination with a data look up table of different threshold intensities for different types of batteries (e.g. per battery part number) or different types of individual cell part numbers ensures the resulting fill level is correct, even between different size cells and accounting for vent access.

[0028] In a second example, an embodiment of the systematic automatic watering system uses a non-contact sonar signal to bounce off the top of the water. The analog output is based off the time it takes for the sound to return to the sensor. Again, the use of an acoustic sensor in combination with a database of different sound/time thresholds for different batteries or cells (e.g. per battery part number or individual cell part number) to ensure the liquid level is correct between different size cells and accounting for vent access.

[0029] It is important to note that the electrolyte (e.g. potassium hydroxide) does not come in contact with the probe (gun). Consequently, a much more reliable level of electrolyte filling is provided by the system tool and methodology, and a much longer longevity of the tool is provided.

[0030] Turning now to FIG. 1, an illustration of a block diagram of a systematic automatic watering system tool **100** is depicted in accordance with an illustrative embodiment. The phrase systematic automatic watering can be abbreviated with the acronym SAW. The systematic automatic watering system tool **100** comprises a non-contact electrolyte level sensing probe **110**. The non-contact electrolyte levels sensing probe **110** includes an insertion stop **120** the non-contact electrolyte level sensing probe **110** includes a contact switch **130**.

[0031] The non-contact electrolyte level sensing probe **110** includes an insertion probe **140**. The insertion probe **140** includes a fill conduit **150**. The insertion probe **140** includes a signal transmitting transducer **160**. The insertion probe **140** includes a signal receiving transducer **170**. The signal transmitting transducer can include an acoustic transmitter and/or an optical transmitter and the signal receiving transducer can include an acoustic receiver and/or an optical receiver.

[0032] This systematic automatic watering system tool **100** includes a control panel **180**. The control panel **180** includes a flow rate display **183**. The control panel **180** includes a stop flow display light **187**. An aircraft battery servicing machine can include the insertion probe and/or the control panel.

[0033] Referring to FIG. 2, a non-contact electrolyte level sensing probe **200** of a systematic automatic watering system tool is shown. Non-contact electrolyte level sensing probe **200** includes an electrolyte supply hose **210** and a signal transmitting cable **220**. The non-contact electrolyte level sensing probe **200** includes a signal receiving cable

230. The non-contact electrolyte level sensing probe **200** includes a contact switch signal cable **235**. Of course, embodiments of this disclosure are not limited to the depicted configuration of hose and cables. In particular, alternative embodiments can combine the signal receiving cable and the signal transmitting cable into a single cable by separating the transmitting signal and the receiving signal across multiple conductors within the single cable or by signal separation on a single circuit.

[0034] The non-contact electrolyte level sensing probe **200** is advantageously configured to have an ergonomically shaped handle **240**. The non-contact electrolyte level sensing probe **200** includes an insertion stop **250** configured to make contact with a flange **270** of an electrolyte fill port **275** of a battery cell. The insertion stop **250** includes a contact switch **260**. The contact switch **260** is configured to be actuated by being depressed by contact with flange **270** of electrolyte fill port **275**. This actuation can send a signal to a control panel that the non-contact electrolyte level sensing probe **200** is located at a fixed location relative to flange **270** of electrolyte fill port **275**. Thus, the tip of the non-contact electrolyte level sensing probe **200** will be a fixed distance beneath the flange **270**.

[0035] The non-contact electrolyte level sensing probe **200** includes an insertion tip **280**. The insertion tip **280** is configured to present an open end of an insertion conduit **285** within the battery cell. The insertion tip **280** includes and the insertion conduit **285** at least partially surrounds an electrolyte supply tip **290**. The electrolyte supply tip **290** is advantageously configured to include an angular orifice **291** to help direct the flow of electrolyte fluid.

[0036] The insertion tip **280** includes and the insertion conduit **285** at least partially surrounds a signal transmitting transducer **293**. The insertion tip **280** includes and the insertion conduit **285** at least partially surrounds a signal receiving transducer **295**. Of course, embodiments of this disclosure are not limited to the depicted configuration of components. In particular, the electrolyte supply tip, the signal transmitting transducer, and the signal receiver can be rearranged or even combined in fewer, or in a unitary structure. Also, there can be additional subcomponents located on the insertion conduit such as additional transducers, sensors and other subcomponents.

[0037] Referring to FIG. 3, a control panel **300** of a systematic automatic watering system tool is shown. The control panel **300** includes a touch screen **310**. The control panel **300** includes a flow stopped indicator **320**. The flow stopped indicator **320** can include a large red light.

[0038] The touch screen **310** includes a flow rate indicator **330**. The touch screen **310** includes a reservoir level indicator **340**. Of course, the invention is not limited to the control panel or touch screen of this embodiment, and alternative embodiments can include alternative control panels and/or alternative touch screens with alternative controls.

[0039] Referring to FIG. 4, a process for systematic automatic battery watering is shown. Block **410** includes providing a systematic automatic battery watering system tool comprising a non-contact electrolyte level sensing probe comprising an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port, and an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer. Block **420** includes introducing the insertion tip into the battery cell electrolyte fill port. Block **430** includes abutting the insertion stop against

the edge of the battery cell electrolyte fill port. Block **440** includes starting a flow of a liquid through the fill conduit and into the battery cell. Block **450** includes emitting a transmitted signal using the signal transmitting transducer. Block **460** includes monitoring a strength of a received signal reflected by a surface of an electrolyte in the battery cell using the signal receiving transducer. Block **470** includes detecting the strength of the received signal reflected by the surface of the electrolyte in the battery cell exceeding a received signal strength threshold. Block **480** includes stopping the flow of the liquid into the battery cell.

[0040] Referring to FIG. 5, a process for scanning a bar code or QR code located on a battery cell and setting a received signal strength threshold based on a predetermined set point associated with a type of cell characterized by the bar code or QR code located on the battery cell is shown. Block **510** includes, before introducing, setting the received signal strength threshold based on a predetermined set point associated with a type of cell characteristic of the battery cell. Block **520** includes, before setting, scanning a code located on the battery cell.

[0041] Referring to FIG. 6, a process for closing a contact switch, opening the contact switch, and shutting off a flow of liquid is shown. Block **610** includes, after introducing and before starting, closing a contact switch located on the non-contact electrolyte level sensing probe and adapted to operatively cooperate with the battery cell electrolyte fill port when abutting the insertion stop against the edge of the battery cell electrolyte fill port to enable starting the flow of the liquid. Block **620** includes, responsive to opening the contact switch after closing the contact switch, shutting off the flow of liquid.

[0042] Referring to FIG. 7, a process for monitoring a received optical signal using a photoelectric device is shown. Block **710** depicts an aspect of an embodiment where the transmitted signal comprises a transmitted optical signal and wherein the received signal reflected by the surface of the electrolyte in the battery cell comprises a received optical signal. Block **720** depicts an aspect of an embodiment where the transmitted optical signal is emitted from a light emitting diode and the received signal is monitored using a photoelectric device.

[0043] Referring to FIG. 8, a process for monitoring a received acoustic signal using a sonar sensor is shown. Block **810** depicts an aspect of an embodiment where the transmitted signal comprises a transmitted acoustic signal and wherein the received signal reflected by the surface of the electrolyte in the battery cell comprises a received acoustic signal. Block **820** depicts an aspect of an embodiment where the transmitted acoustic signal is emitted from a sonar transducer and the received signal is monitored using sonar sensor.

[0044] Illustrative embodiments of the present disclosure may be described in the context of aircraft manufacturing and service method **900** as shown in FIG. 9 and aircraft **1000** as shown in FIG. 10. Turning first to FIG. 9, an illustration of an aircraft manufacturing and service method in the form of a block diagram is depicted in accordance with an illustrative embodiment. During pre-production, aircraft manufacturing and service method **900** may include specification and design **902** of aircraft **1000** in FIG. 10 and material procurement **904**.

[0045] During production, component and subassembly manufacturing **906** and system integration **908** of aircraft

1000 takes place. Thereafter, aircraft **1000** may go through certification and delivery **910** in order to be placed in service **912**. While in service **912** by a customer, aircraft **1000** is scheduled for routine maintenance and service **914**, which may include modification, reconfiguration, refurbishment, or other maintenance and service.

[0046] Each of the processes of aircraft manufacturing and service method **900** may be performed or carried out by a system integrator, a third party, and/or an operator. In these examples, the operator may be a customer. For the purposes of this description, a system integrator may include, without limitation, any number of aircraft manufacturers and major-system subcontractors; a third party may include, without limitation, any number of vendors, subcontractors, and suppliers; and an operator may be an airline, a leasing company, a military entity, a service organization, and so on.

[0047] With reference now to FIG. **10**, an illustration of an aircraft in a form of a block diagram is depicted in which an illustrative embodiment may be implemented. In this example, aircraft **1000** is produced by aircraft manufacturing and service method **900** of FIG. **9** and may include airframe **1002** with plurality of systems **1004** and interior **1006**. Examples of systems **1004** include one or more of propulsion system **1008**, electrical system **1010**, hydraulic system **1012**, and environmental system **1014**. Any number of other systems may be included.

[0048] Apparatuses and methods embodied herein may be employed during at least one of the stages of aircraft manufacturing and service method **900**. One or more illustrative embodiments may be manufactured or used during at least one of component and subassembly manufacturing **906**, system integration **908**, in service **912**, or maintenance and service **914** of FIG. **9**.

[0049] The description of the different illustrative embodiments has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the embodiments in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different illustrative embodiments may provide different features as compared to other illustrative embodiments. The embodiment or embodiments selected are chosen and described in order to best explain the principles of the embodiments, the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

What is claimed is:

1. A method of automatically watering a battery cell, comprising:

providing a systematic automatic battery watering system tool comprising a non-contact electrolyte level sensing probe comprising an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port, and an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer;

introducing the insertion tip into the battery cell electrolyte fill port;

abutting the insertion stop against the edge of the battery cell electrolyte fill port;

starting a flow of a liquid through the fill conduit and into the battery cell;

emitting a transmitted signal using the signal transmitting transducer;

monitoring a strength of a received signal reflected by a surface of an electrolyte in the battery cell using the signal receiving transducer;

detecting the strength of the received signal reflected by the surface of the electrolyte in the battery cell exceeding a received signal strength threshold; and
stopping the flow of the liquid into the battery cell.

2. The method of claim 1, further comprising, before introducing, setting the received signal strength threshold based on a predetermined set point associated with a type of cell characteristic of the battery cell.

3. The method of claim 2, further comprising, before setting, scanning a code located on the battery cell.

4. The method of claim 1, further comprising, after introducing and before starting, closing a contact switch located on the non-contact electrolyte level sensing probe and adapted to operatively cooperate with the battery cell electrolyte fill port when abutting the insertion stop against the edge of the battery cell electrolyte fill port to enable starting the flow of the liquid.

5. The method of claim 4, further comprising, responsive to opening the contact switch after closing the contact switch, shutting off the flow of liquid.

6. The method of claim 1, wherein the transmitted signal comprises a transmitted optical signal and wherein the received signal reflected by the surface of the electrolyte in the battery cell comprises a received optical signal.

7. The method of claim 6, wherein the transmitted optical signal is emitted from a light emitting diode and the received signal is monitored using a photoelectric device.

8. The method of claim 1, wherein the transmitted signal comprises a transmitted acoustic signal and wherein the received signal reflected by the surface of the electrolyte in the battery cell comprises a received acoustic signal.

9. The method of claim 8, wherein the transmitted acoustic signal is emitted from a sonar transducer and the received signal is monitored using sonar sensor.

10. An aircraft battery comprising a cell watered using the method of claim 6.

11. An apparatus for automatically watering a battery cell, comprising:

a control panel;

a non-contact electrolyte level sensing probe coupled to the control panel, the non-contact electrolyte level sensing probe comprising:

an insertion stop adapted to abut against an edge of a battery cell electrolyte fill port;

an insertion tip comprising a fill conduit;

a signal transmitting transducer; and

a signal receiving transducer adapted to stop a flow of liquid into the battery cell in response to detecting a strength of a received signal reflected by a surface of an electrolyte in the battery cell exceeding a received signal strength threshold.

12. The apparatus of claim 11, wherein the control panel comprises a flow rate display.

13. The apparatus of claim 11, wherein the control panel comprises a stop flow display light.

14. The apparatus of claim 11, further comprising a contact switch located on the non-contact electrolyte level sensing probe and adapted to operatively cooperate with a battery cell electrolyte fill port when abutting the insertion stop against the edge of the battery cell electrolyte fill port to enable starting the flow of the liquid.

15. The apparatus of claim **11**, wherein the signal transmitting transducer comprises an optical transmitter and the signal receiving transducer comprises an optical receiver.

16. The apparatus of claim **15**, wherein the optical transmitter comprises a light emitting diode and the optical receiver comprises a photoelectric device.

17. The apparatus of claim **11**, wherein the signal transmitting transducer comprises an acoustic transmitter and the signal receiving transducer comprises an acoustic receiver.

18. The apparatus of claim **17**, wherein the acoustic transmitter comprises a sonar transducer and the acoustic receiver comprises a sonar sensor.

19. An aircraft battery servicing machine comprising the apparatus of claim **11**.

20. A method of automatically watering, comprising:

providing a systematic automatic watering system tool comprising a non-contact liquid level sensing probe comprising an insertion stop adapted to abut against an edge of a cell fill port, and an insertion tip comprising a fill conduit, a signal transmitting transducer, and a signal receiving transducer;

setting a received signal strength threshold based on a predetermined set point associated with a type of cell characteristic of a cell;

introducing the insertion tip into the cell fill port of the cell;

closing a contact switch located on the non-contact liquid level sensing probe and adapted to operatively cooperate with the cell fill port when abutting the insertion stop against the edge of the cell fill port to enable starting a flow of a liquid;

abutting the insertion stop against the edge of the cell fill port;

starting a flow of a liquid through the fill conduit and into the cell;

emitting a transmitted signal using the signal transmitting transducer;

monitoring a strength of a received signal reflected by a surface of a liquid in the cell using the signal receiving transducer;

detecting the strength of the received signal reflected by the surface of the liquid in the cell exceeding the received signal strength threshold; and

stopping the flow of the liquid into the cell.

* * * * *